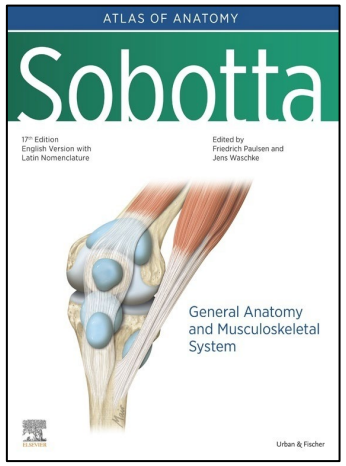
Modul 5.3 Bevægeapparatet

Overekstremitetens (UE) - regional anatomi





Tong Tong
Assistant Professor

Normal text = Danish
Normal text + Italic = Latin anatomical term
Italic + smaller + green = English

Fra systematisk til regional anatomi

From systematic to regional anatomy

Systematisk anatomi: Hvad er strukturerne?
Systematic anatomy: What are the structures?

Regional anatomi: Hvor ligger strukturerne i forhold til hinanden?
Regional anatomy: Where are structures located relative to one another?

- Muskler bruges som grænser og landemærker
  Muscles are used as boundaries and landmarks
- Kar og nerver følges gennem regioner
  Vessels and nerves are traced through regions

Læringsmål:

- **Redegøre for de vigtigste anatomiske forhold i *regio deltoidea*, herunder centrale overfladiske og dybe strukturer**
  Describe the key anatomical relationships in the deltoid region, including important superficial and deep structures

- **Redegøre for afgrænsninger og indhold i *regio axillaris*, herunder de vigtigste kar, nerver og lymfatiske strukturer**
  Describe the boundaries and contents of the axillary region, including the major vessels, nerves and lymphatic structures

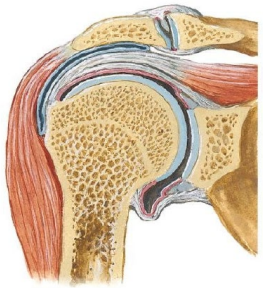
- **Redegøre for de vigtigste regionale relationer i *regio brachialis*, herunder muskelloger samt forløbet af større kar og perifere nerver**
  Describe the major regional relationships in the brachial region, including the muscular compartments and the course of major vessels and peripheral nerves

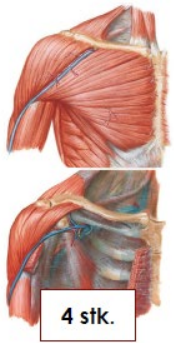
- **Redegøre for afgrænsninger, gulv, loft og indhold i *regio cubitalis* / *fossa cubitalis***
  Describe the boundaries, floor, roof and contents of the cubital region / cubital fossa

- **Redegøre for de vigtigste anatomiske passager og relationer i *regio carpalis*, herunder *canalis carpi* og tabatièren**
  Describe the major anatomical passageways and relationships in the carpal region, including the carpal tunnel and anatomical snuffbox

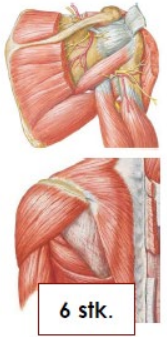
Skulderen
The shoulder





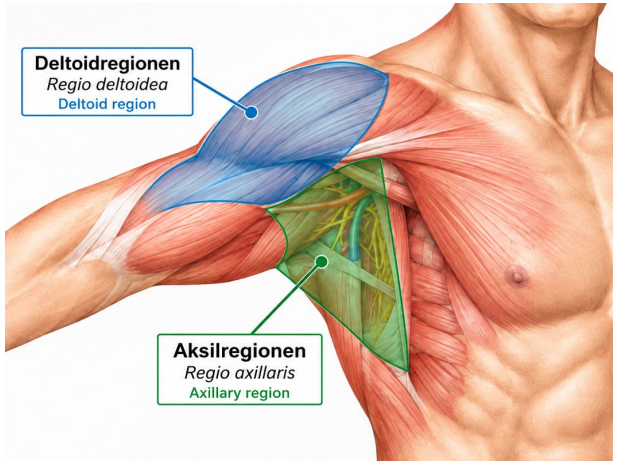






Deltoidregionen
Regio deltoidea
Deltoid region

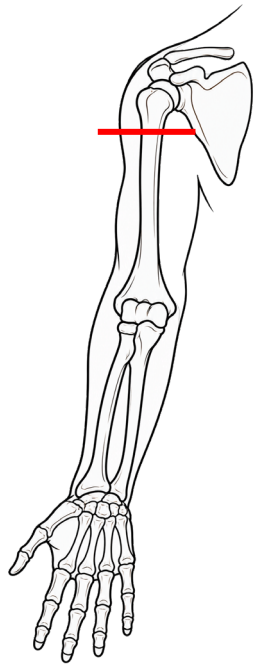
- Lateral skulderregion
  Lateral shoulder region
- *M. deltoideus*, *n. axillaris*, proximal humerus
  Deltoid, axillary nerve, proximal humerus
- Klinik: deltoidsvaghed, skade på n. axillaris, proximal humerusfraktur og intramuskulær injektion
  Clinical: deltoid weakness, axillary nerve injury, proximal humeral fracture and intramuscular injection



Aksilregionen
Regio axillaris
Axillary region

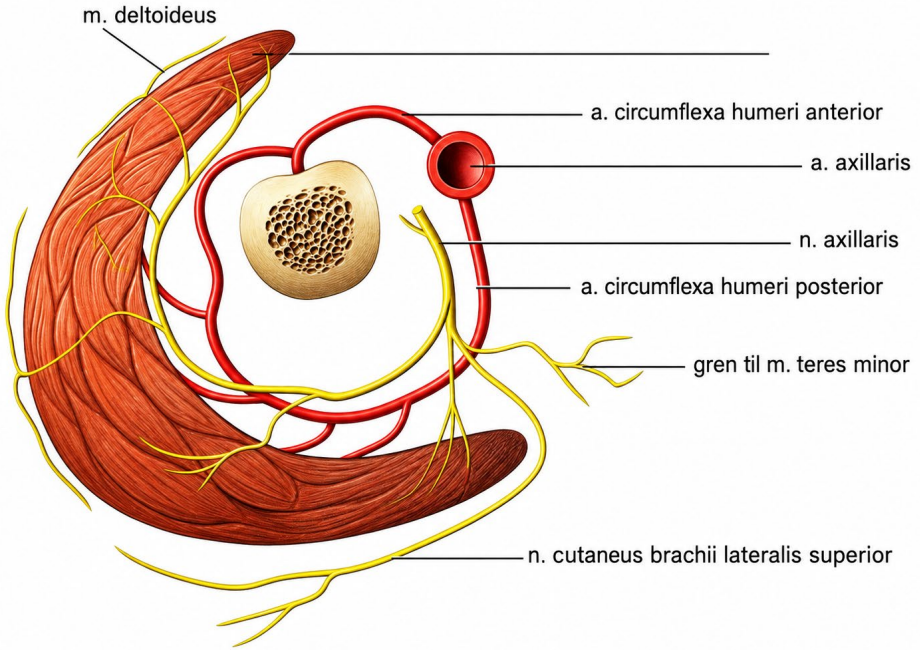
- Passage mellem thorax og arm
  Passage between thorax and arm
- A./v. axillaris, plexus brachialis, lymfeknuder
  Axillary vessels, brachial plexus, lymph nodes
- Klinisk: lymfeknudeundersøgelse, neurovaskulær skade, kompression og aksillær kirurgi
  Clinically relevant in lymph-node examination, neurovascular injury, compression and axillary surgery

Deltoidregionen
Regio deltoidea
Deltoid region



- Dækker den laterale del af skulderen omkring *m. deltoideus*
  Covers the lateral part of the shoulder around the deltoid muscle

- Vigtige dybe relationer:
  Important deep relationships:
  - *N. axillaris*
    Axillary nerve
  - *A./v. circumflexa humeri posterior*
    Posterior circumflex humeral vessels
  - Proximal humerus og skulderled
    Proximal humerus and shoulder joint



- Klinisk betydning
  - Skade på *n. axillaris*
    Axillary nerve injury
  - Fraktur omkring *collum chirurgicum humeri*
    Fracture around the surgical neck of the humerus
  - Intramuskulær injektion i *m. deltoideus*
    Intramuscular injection into the deltoid muscle

Aksilregionen
Regio axillaris
Axillary region

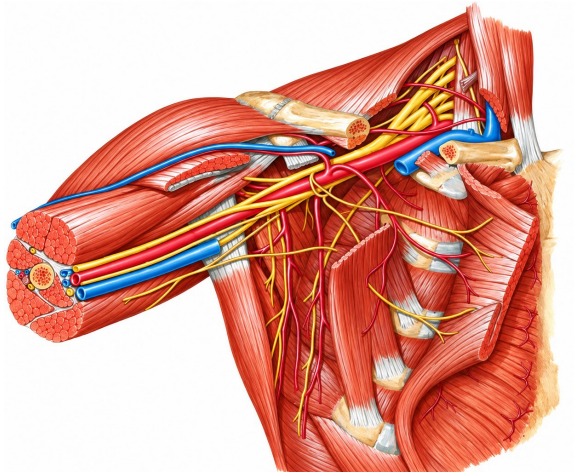
Aksillen (armhulen)
Axilla (armpit)



- **Stor bindevævsrum**
  Large connective-tissue space

- Afgrænses af forreste og bageste aksilfold
  Bounded by the anterior and posterior axillary folds

- **Indeholder**
  - ***A. axillaris + v. axillaris***
    Axillary artery and axillary vein
  - ***Plexus brachialis***
    Brachial plexus
  - ***Lymphonodi axillares***
    Axillary lymph nodes
  - Kutane nerver
    Cutaneous nerves
  - *Processus axillaris mammae*
    Axillary process of the breast / axillary tail of the breast
  - Fedtvæv
    Adipose tissue / fat



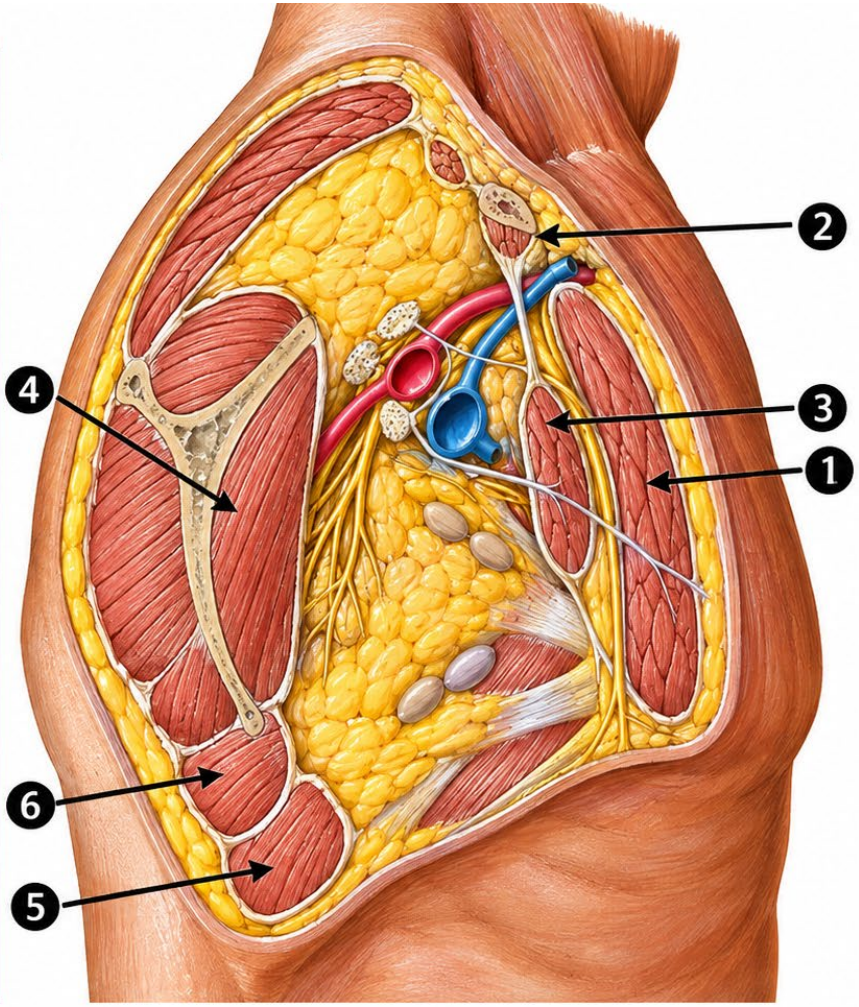
Aksilregionen
Regio axillaris
Axillary region

Afgrænsninger

Forvæg
- ❶ M. pectoralis major
- ❷ M. subclavius
- ❸ M. pectoralis minor

Bagvæg
- ❹ M. subscapularis
- Tilskud fra:
- ❺ M. latissimus dorsi
- ❻ M. teres major

Medialvæg
- M. serratus anterior
- Costa + intercostalmuskler



Aksilregionen
Regio axillaris
Axillary region

A. axillaris
Axillary artery

Fortsættelse af *a. subclavia*
Continuation of the subclavian artery

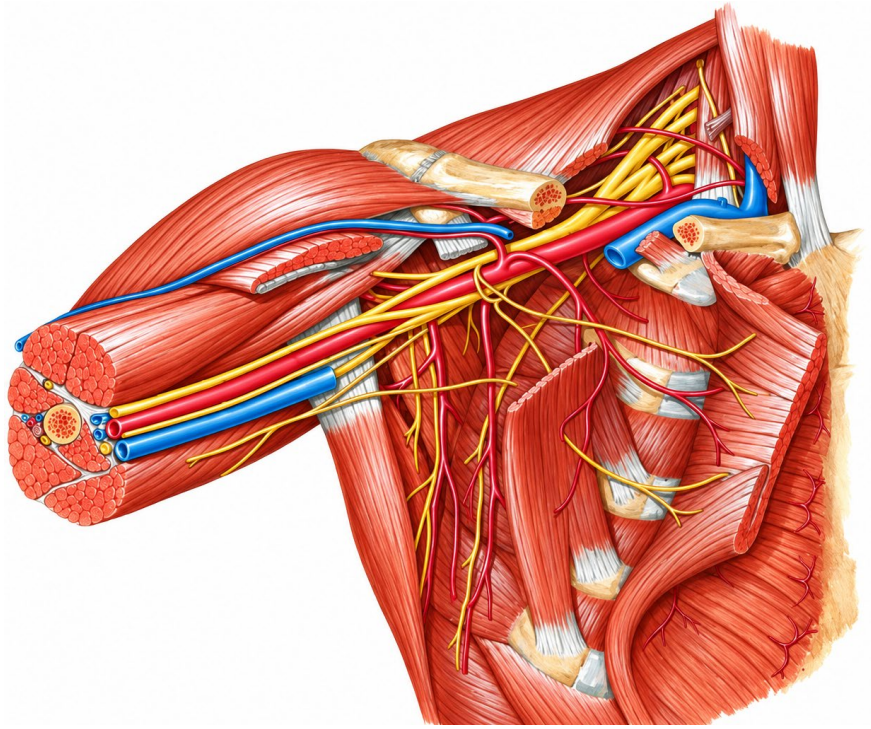
Løber i *fossa axillaris* efter costa I
Runs in the axilla after crossing the first rib

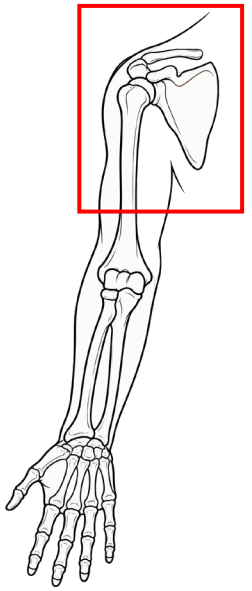
Ligger i tæt relation til *plexus brachialis*
Lies in close relation to the brachial plexus

V. axillaris ligger medialt
The axillary vein lies medially

Fortsætter som *a. brachialis*
Continues as the brachial artery

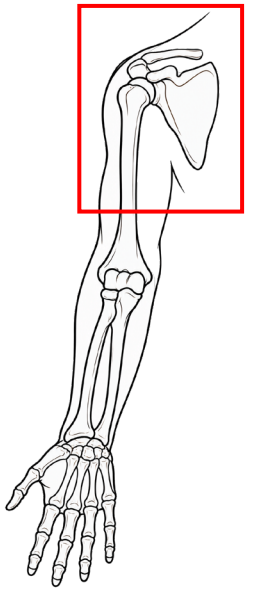
Overgangen sker ved den bageste aksilfold
The transition occurs at the posterior axillary fold



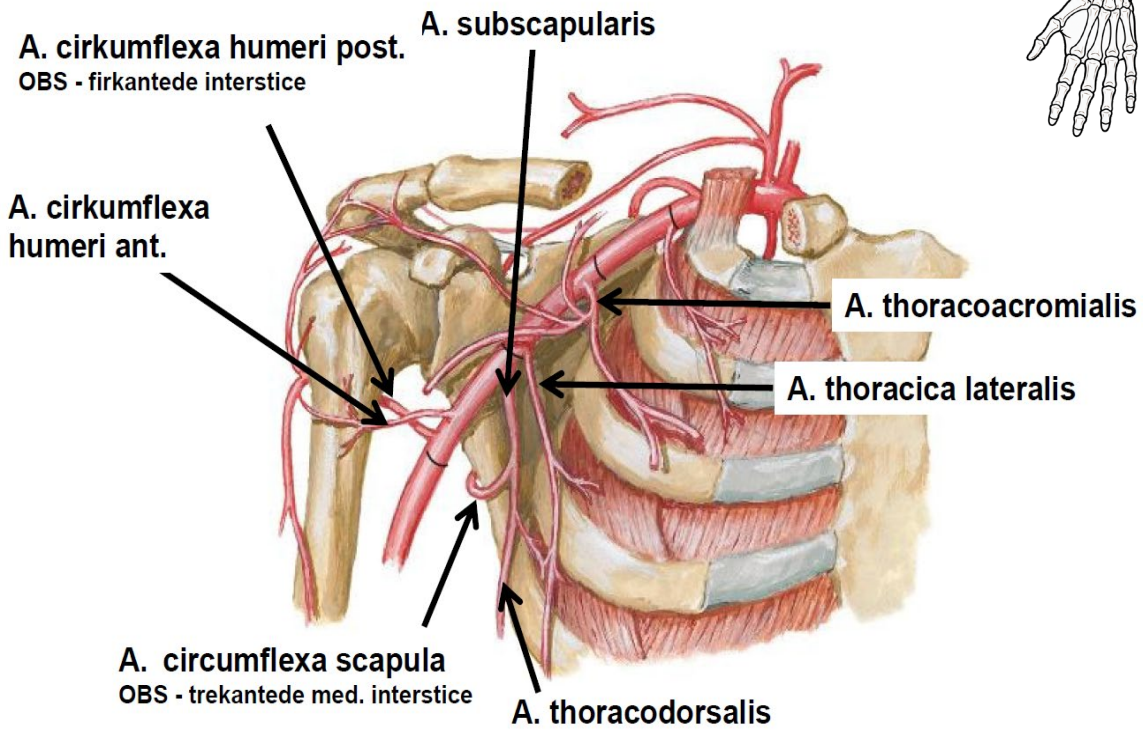


Aksilregionen
Regio axillaris
Axillary region

A. axillaris
Axillary artery

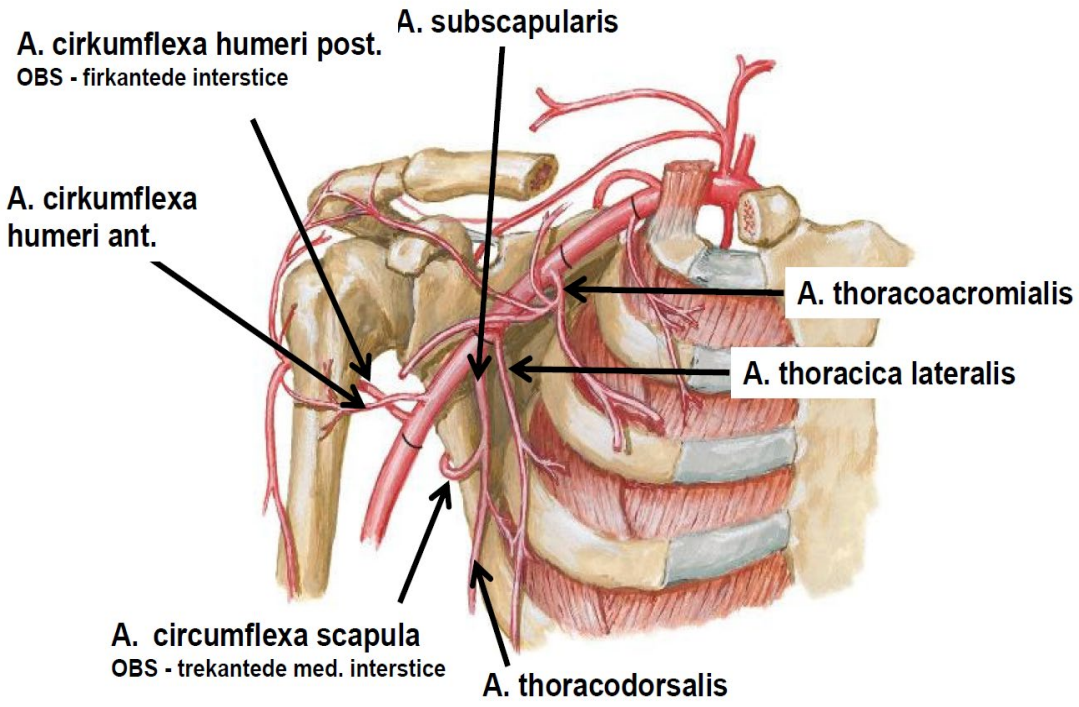


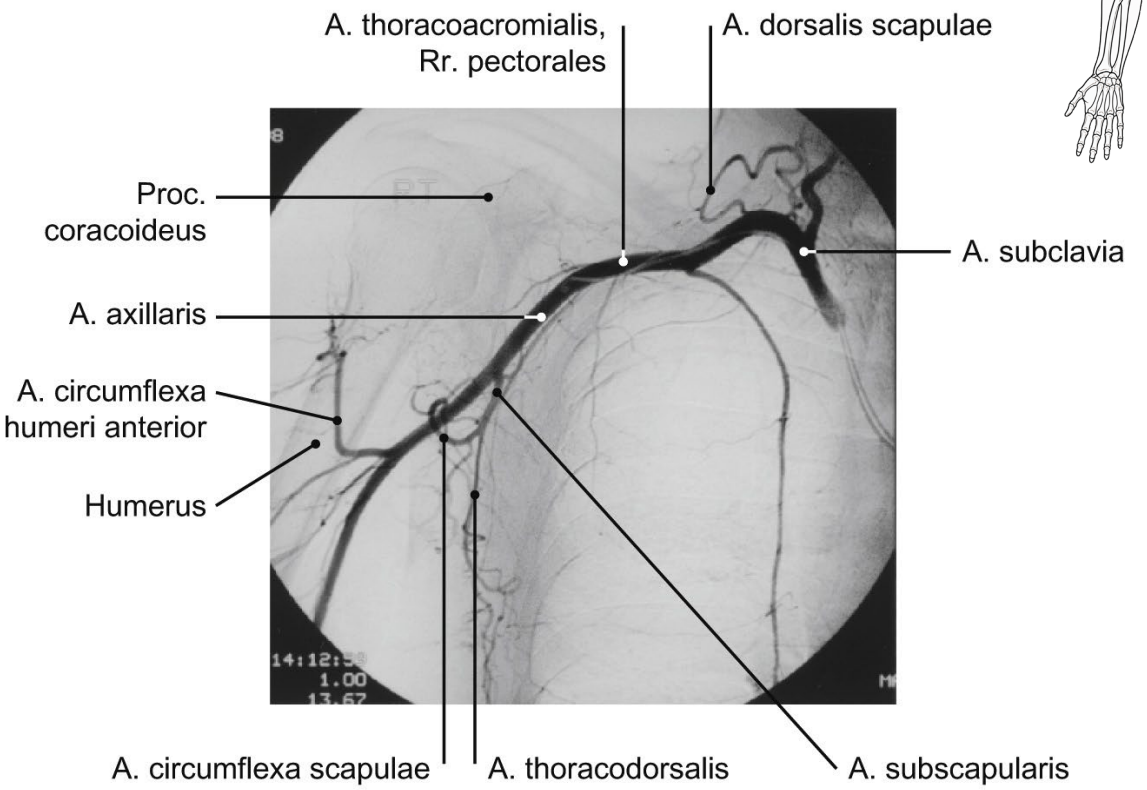
- *A. circumflexa humeri anterior*
  Anterior circumflex humeral artery
- *A. circumflexa humeri posterior*
  Posterior circumflex humeral artery
  OBS – firkantede interstice
  Note – passes through the quadrangular space
- *A. subscapularis*
  Subscapular artery
- *A. thoracoacromialis*
  Thoracoacromial artery
- *A. thoracica lateralis*
  Lateral thoracic artery
- *A. circumflexa scapulae*
  Circumflex scapular artery
  OBS – trekantede mediale interstice
  Note – passes through the triangular space
- *A. thoracodorsalis*
  Thoracodorsal artery

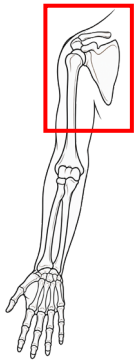


Aksilregionen
Regio axillaris
Axillary region

A. axillaris
Axillary artery

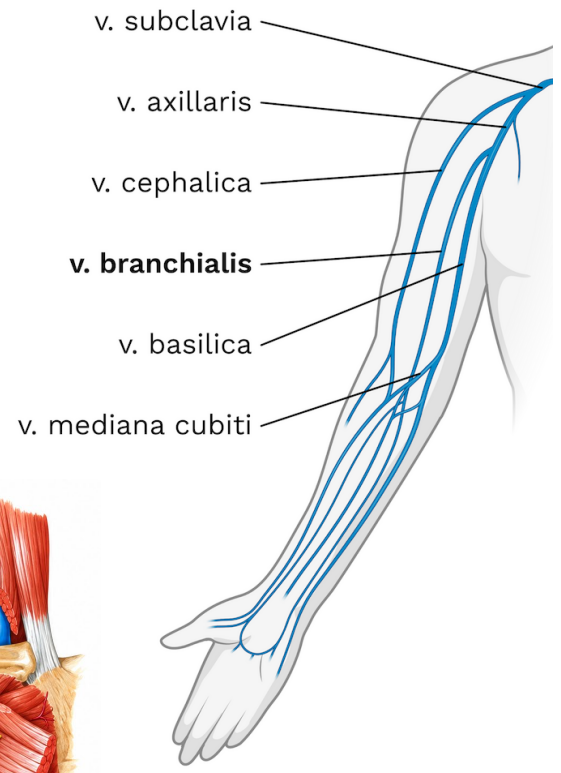




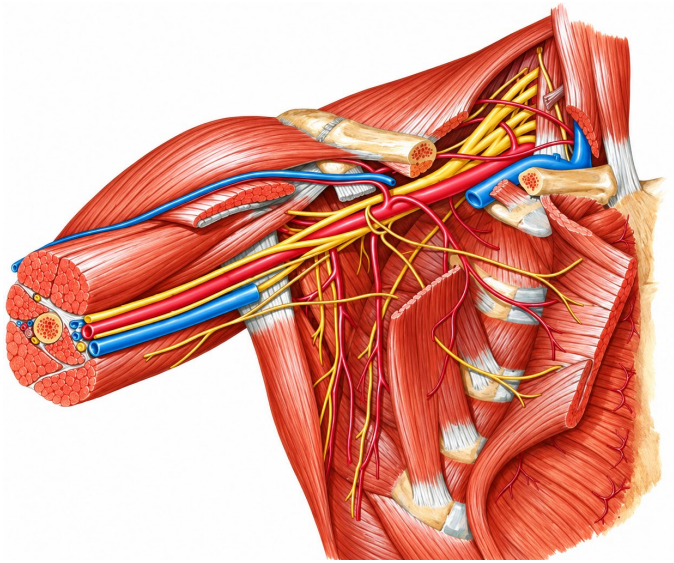


Aksilregionen
Regio axillaris
Axillary region

Overekstremitetens vener
Veins of the upper limb



- Alt dræneres slutteligt til v. axillaris → v. subclavia
  All venous blood ultimately drains into the axillary vein → subclavian vein
- Profunde vener
  Deep veins
  Følger arterierne
  Accompany the arteries
- Superficielle vener
  Superficial veins
- Synlige under huden
  Visible beneath the skin
- Veneklapper
  Venous valves
  Hjælper med at sikre blodets strømning mod hjertet
  Help maintain blood flow toward the heart



Aksilregionen
Regio axillaris
Axillary region

Overekstremitetens lymfedrænage
Lymphatic drainage of the upper limb

Lnn. axillares
Axillary lymph nodes

- **Lateral gruppe**
  Modtager fra ulnar + profund del af OE
  Receives lymph from the ulnar and deep part of the upper extremity

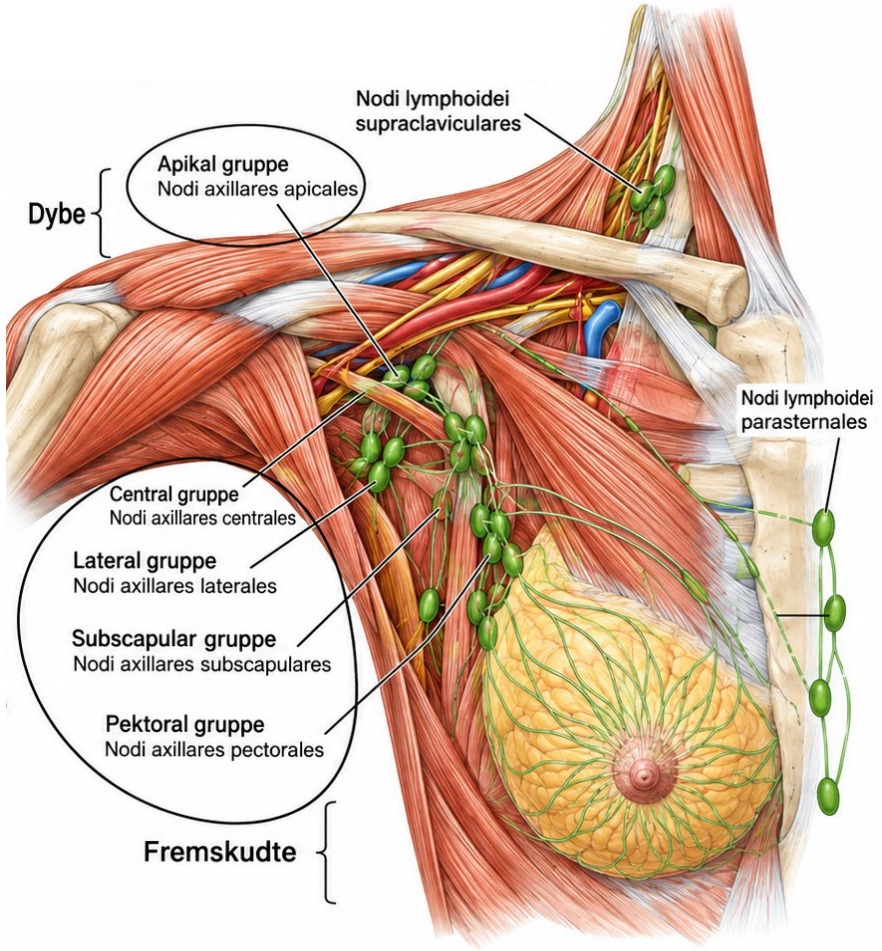
- **Subscapular gruppe**
  Fra skulderregion, øverste del af ryggen
  From the shoulder region and the upper part of the back

- **Pektoral gruppe**
  Fra mamma, forrest kropsvæg over umbilicus
  From the breast and the anterior body wall above the umbilicus

- **Central gruppe**
  Modtager fra de tre ovenstående
  Receives lymph from the three groups above

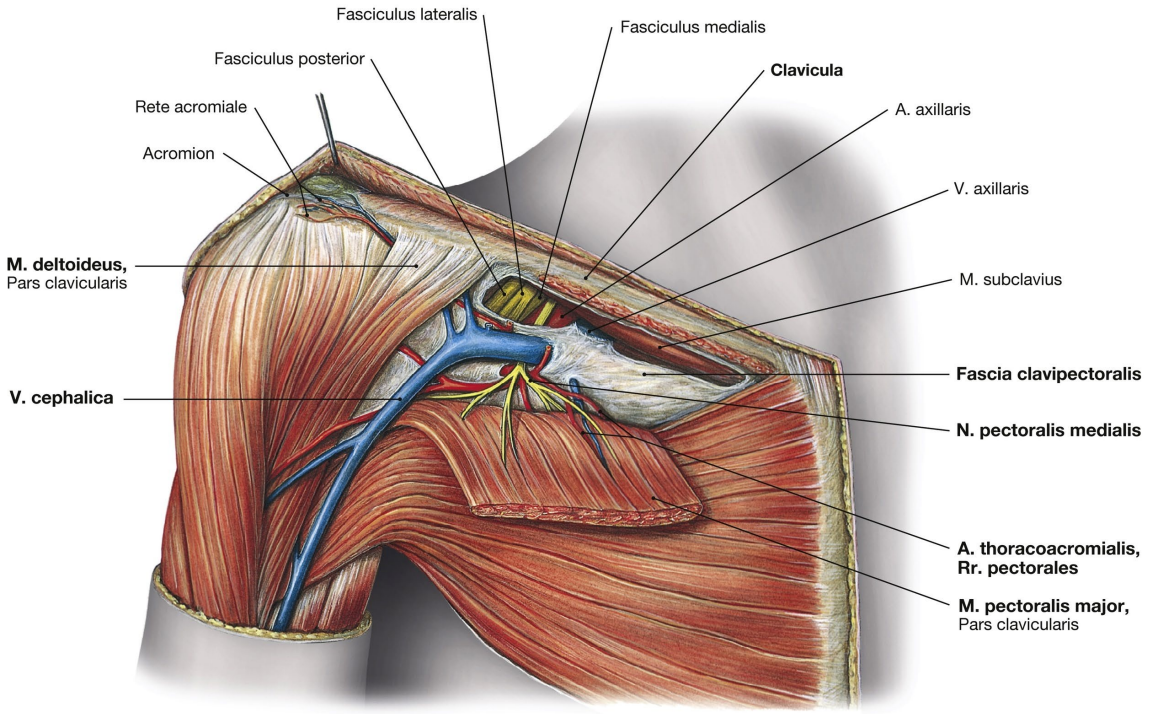
- **Apikal gruppe**
  Modtager fra central gruppe
  Receives lymph from the central group
  Direkte: radial del af hånden (langs *v. cephalica*) og øverste del af mamma
  Also receives directly from the radial part of the hand (along the cephalic vein) and the upper part of the breast

Tømmes i *truncus subclavius* → *ductus thoracicus* (ve.) eller *ductus lymphaticus dexter* → halsvener
Drain into the subclavian trunk → the thoracic duct (left) or the right lymphatic duct → the neck veins



Aksilregionen
Regio axillaris
Axillary region

Trigonum clavipectorale
Clavipectoral triangle



V. cephalica munder i v. axillaris
The cephalic vein drains into the axillary vein

Apikale aksillære lymfeknuder findes i området
Apical axillary lymph nodes are located in the region
→ **klinisk relevant ved brystkræft og lymfespredning**
clinically relevant in breast cancer and lymphatic spread

***Thoracoacromialis* passerer i regionen**
The thoracoacromial artery passes through the region

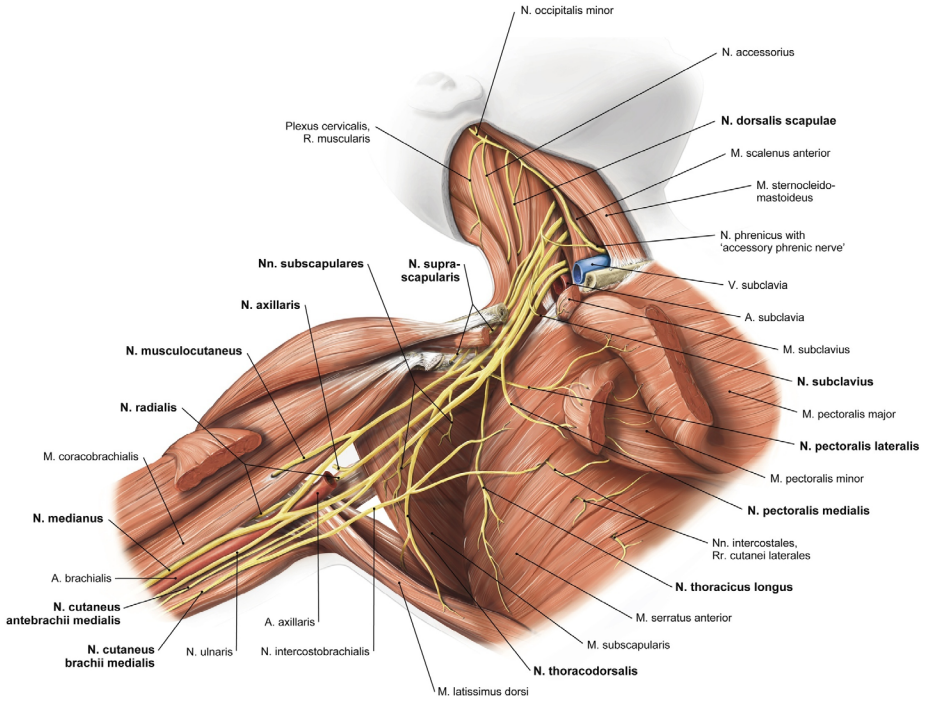
***Nn. pectorales medialis et lateralis* forsyner pectoralmusklerne**
The medial and lateral pectoral nerves supply the pectoral muscles
→ nerveskade kan give muskelsvaghed
nerve injury may cause muscle weakness

***Processus coracoideus* og det deltopectorale interval er vigtige landemærker**
The coracoid process and deltopectoral interval are important landmarks
→ bruges ved anterior kirurgisk adgang til skulderen
used in anterior surgical approaches to the shoulder

Trigonum clavipectorale er et vigtigt kirurgisk, vaskulært og lymfatisk område.
The clavipectoral triangle is an important surgical, vascular and lymphatic region.

Aksilregionen
Regio axillaris
Axillary region

Plexus brachialis
Brachial plexus



Plexet forsyner
The plexus supplies

- Alle muskler på armen
  All muscles of the upper limb → Motoriske fibre
- Hele hudoverfladen på armen
  The skin of the upper limb → Sensoriske fibre
- Sympatisk-innerverede strukturer på armen
  Sympathetically innervated structures of the upper limb → Sympatiske fibre
  - Svedkirtler
    Sweat glands
  - Alle kar
    Blood vessels
  - Glat muskulatur i huden
    Smooth muscle in the skin

Plexus brachialis består af blandede nervegrene
The brachial plexus consists of mixed nerve branches

Plexus = fletværk af nerver
Plexus = a network of nerves

Aksilregionen
Regio axillaris
Axillary region

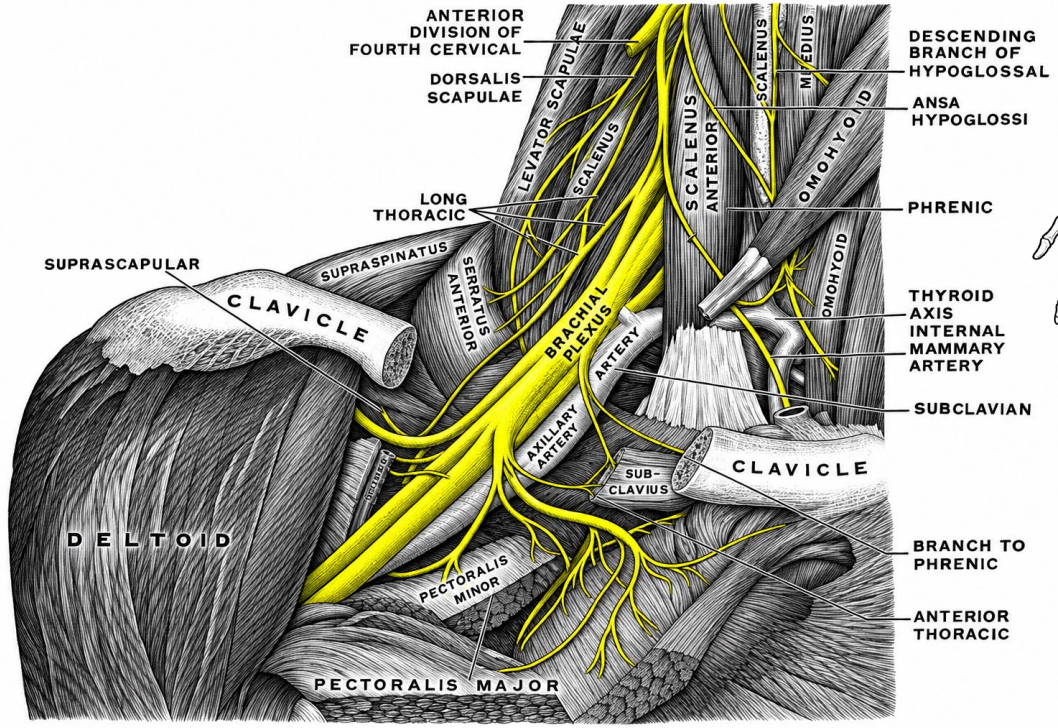
Plexus brachialis
Brachial plexus

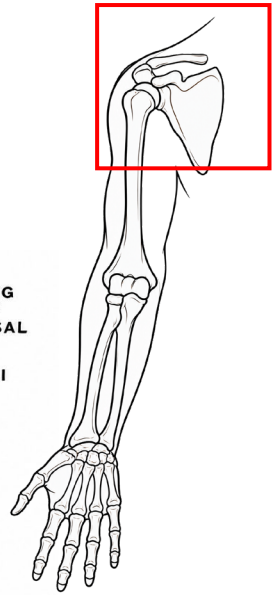
Dannes af de anteriore grene fra C5–T1
Formed by the anterior rami of C5–T1

Passerer gennem scalenerporten (*Spatium interscalenum*)
Passes through the interscalene space

Mellem m. scalenus anterior og m. scalenus medius
Between the anterior and middle scalene muscles

Løber bag clavicula og fortsætter til aksillen
Clavicula → axilla
Passes posterior to the clavicle and continues into the axilla





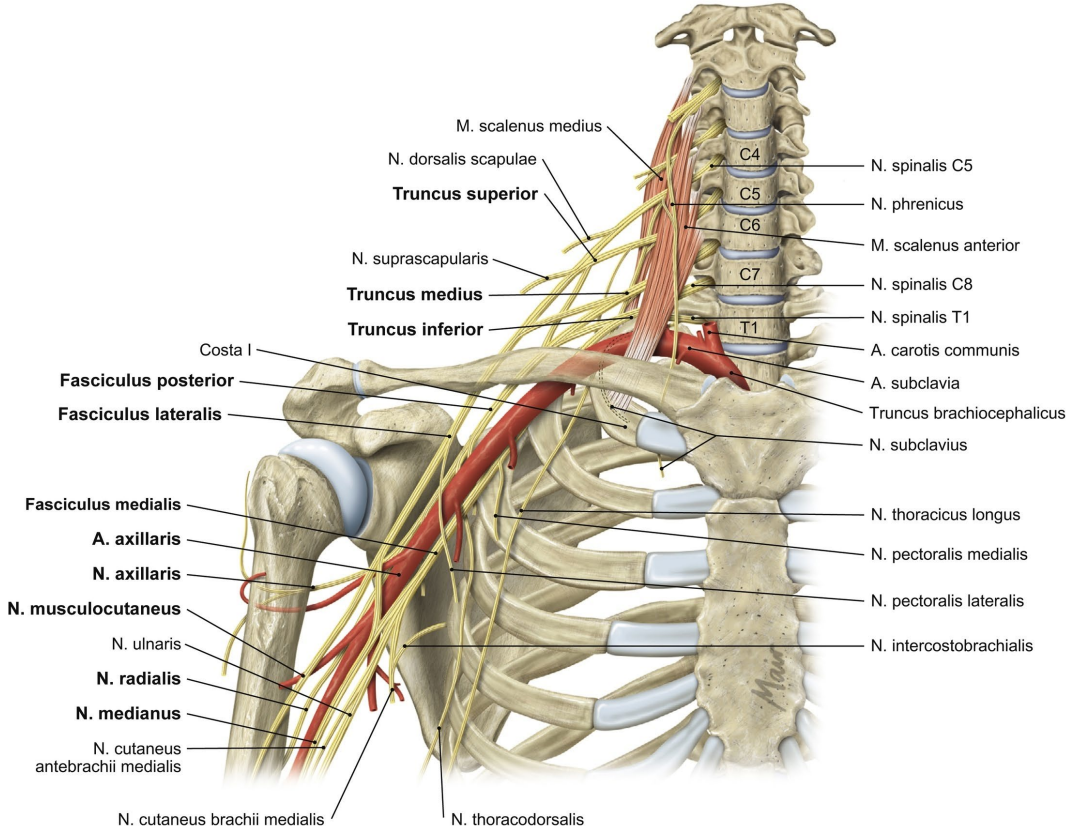
Aksilregionen
Regio axillaris
Axillary region

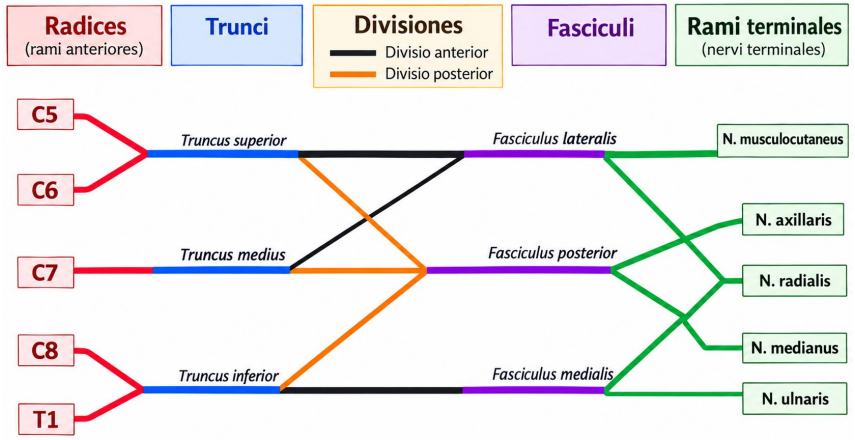
Organisation af *plexus brachialis*
Organization of the brachial plexus

C5–T1
↓
Rødder
Radices / anterior rami
Roots
↓
Trunci
Truncus superior, medius et inferior
Upper, middle and lower trunks
↓
Delinger
Divisiones anteriores et posteriores
Anterior and posterior divisions
↓
Fasciculi
- *Fasciculus lateralis*— *Lateral cord*
- *Fasciculus posterior*— *Posterior cord*
- *Fasciculus medialis*— *Medial cord*

↓
Terminale nerver
Rami terminales
Terminal branches

Plexus brachialis
Brachial plexus



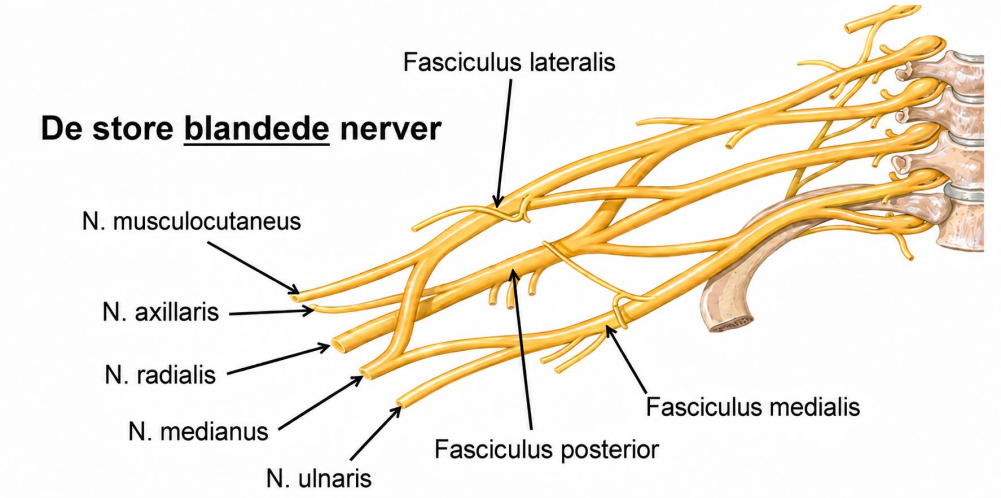


Aksilregionen
Regio axillaris
Axillary region

Plexus brachialis
Brachial plexus

Blanding af nervefibre
Mixing of nerve fibers

- Fibre fra forskellige rygmarvssegmenter samles og fordeles igen
  Fibers from different spinal cord segments converge and are redistributed
- En perifer nerve indeholder derfor fibre fra flere spinalsegmenter
  A peripheral nerve therefore contains fibers from several spinal segments
- De anteriore og posteriore divisioner fordeler fibrene mod fleksor- og ekstensorsiden
  Anterior and posterior divisions redistribute fibers toward flexor and extensor compartments



De fem vigtigste terminale nerver
Five major terminal nerves

- ***N. musculocutaneus***
  Musculocutaneous nerve
- ***N. axillaris***
  Axillary nerve
- ***N. radialis***
  Radial nerve
- ***N. medianus***
  Median nerve
- ***N. ulnaris***
  Ulnar nerve

Brachium (overarmen)
Upper arm

Udspringer fra **fasciculus lateralis**
Arises from the lateral cord
Ligger derfor lateralt for **a. axillaris**
Therefore lies lateral to the axillary artery

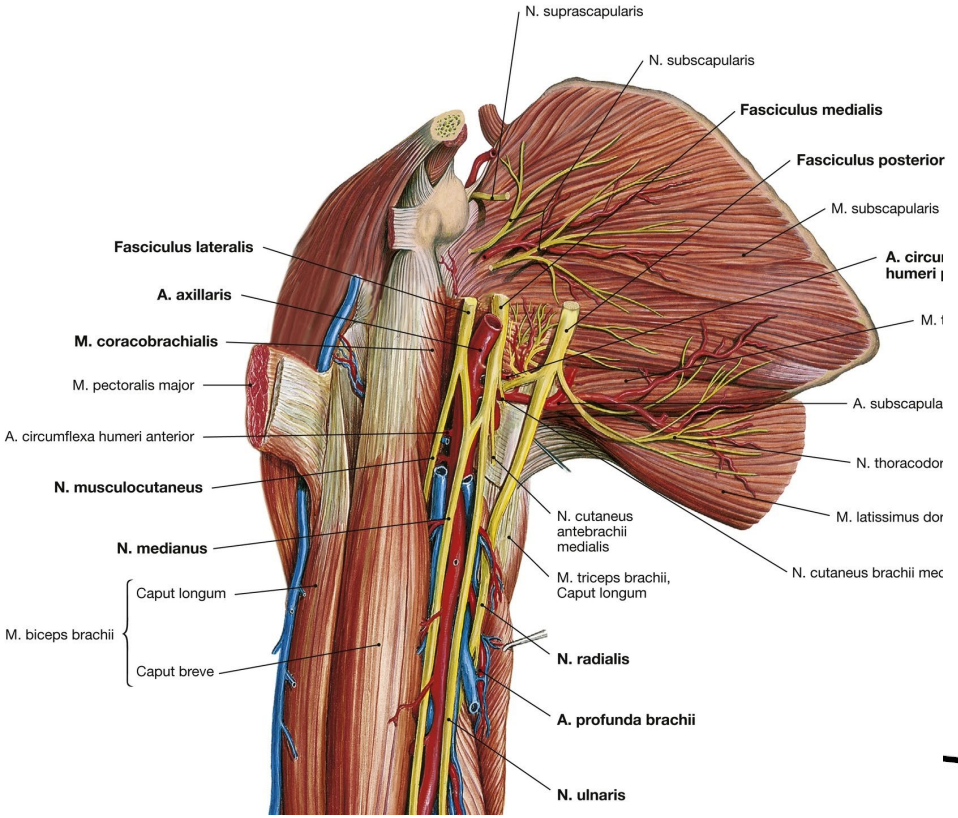
Forløb:
Perforerer **m. coracobrachialis**
Pierces the coracobrachialis muscle
Bliver kutan lateralt for **fossa cubitalis**
Becomes cutaneous lateral to the cubital fossa

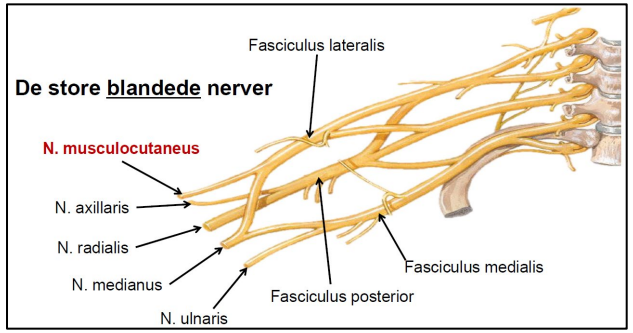
Muskler:
Innerverer fleksorerne i **brachium**
Innervates the flexor muscles of the arm

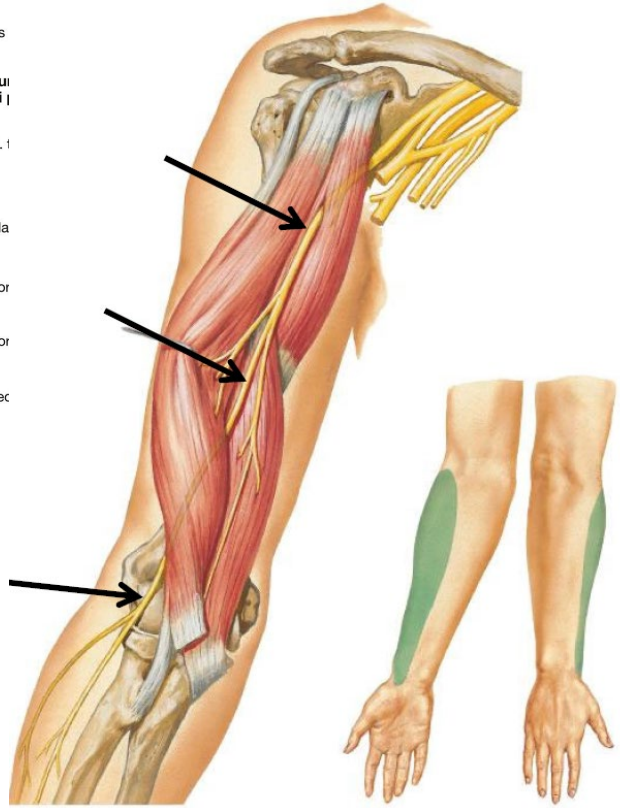
- *musculus coracobrachialis*
- *musculus biceps brachii*
- *musculus brachialis*

Kutant:
Forsyner den laterale del af **underarmen**
Supplies the lateral part of the forearm

N. musculocutaneus
Musculocutaneous nerve

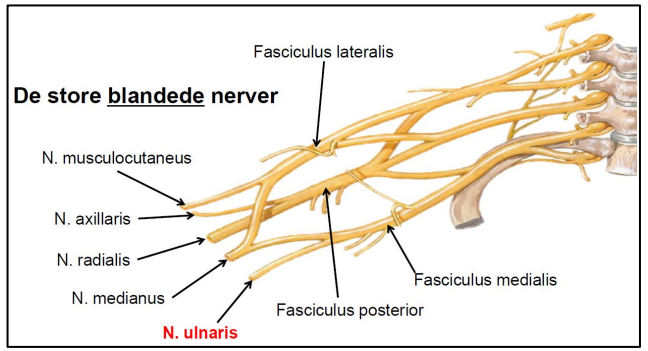






Brachium (overarmen)
Upper arm

N. ulnaris
Ulnar nerve

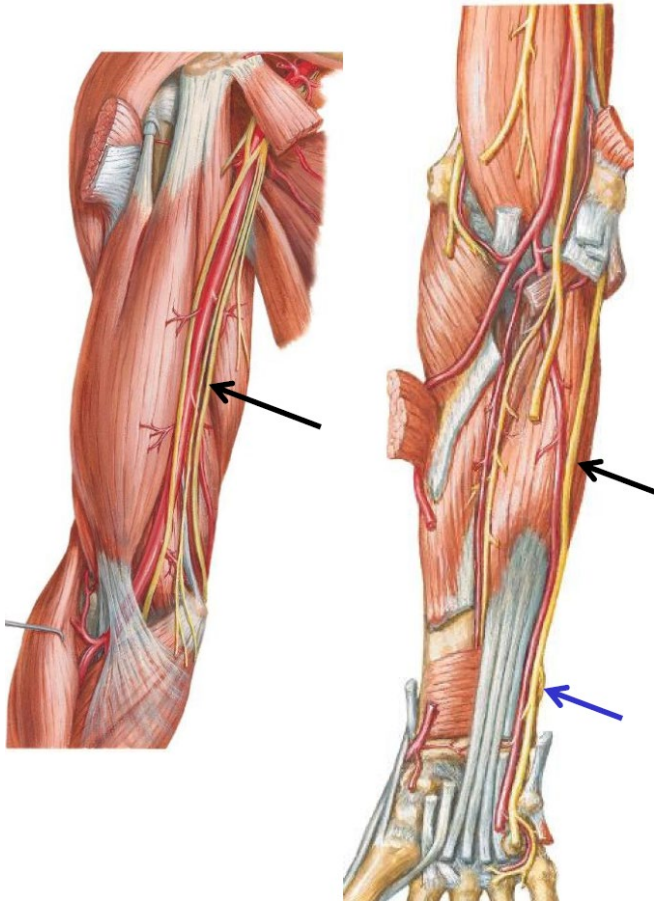


Udspringer fra fasciculus medialis
Arises from the medial cord

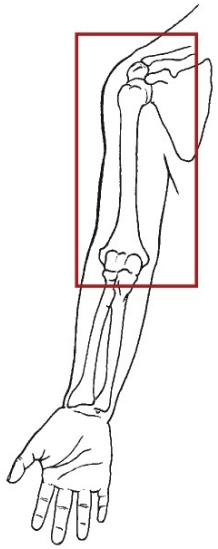
Forløb:
- Langs a. brachialis i sulcus bicipitalis medialis
  Runs along the brachial artery in the medial bicipital groove
- Går om i ekstensorlogen og sulcus n. ulnaris
  Passes into the extensor compartment and into the groove for the ulnar nerve
  Ikke i fossa cubitalis
  Does not pass through the cubital fossa
- Går ind under m. flexor carpi ulnaris
  Passes beneath the flexor carpi ulnaris muscle

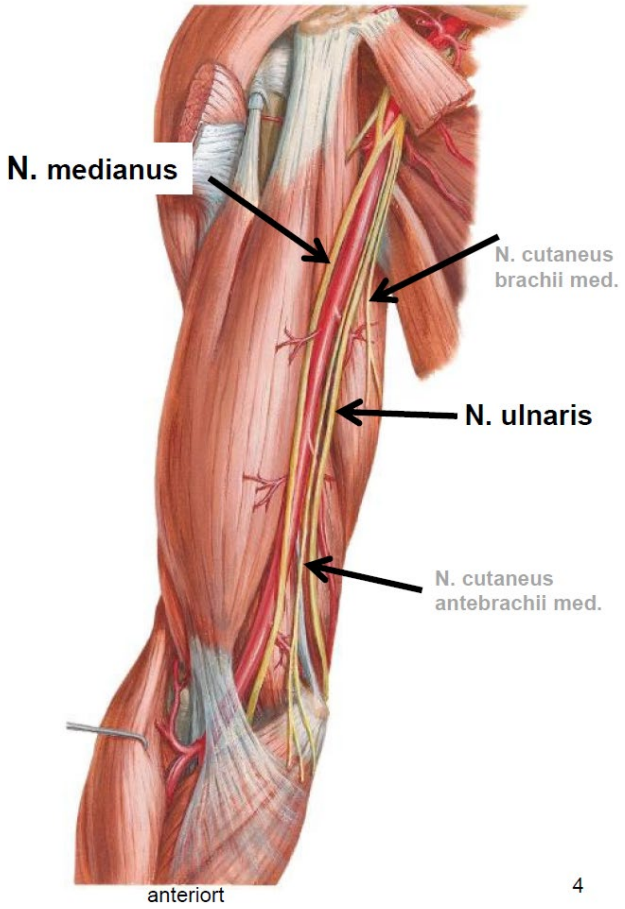
Innerverer m. flexor carpi ulnaris og den ulnare del af m. flexor digitorum profundus
Innervates the flexor carpi ulnaris and the ulnar part of the flexor digitorum profundus

Afgiver ramus dorsalis til håndryggen
Gives off a dorsal branch to the back of the hand



Brachium (overarmen)
Upper arm





A. brachialis
Brachial artery

Fortsættelse af *a. axillaris*
Continuation of the axillary artery

Løber langs humerus i *sulcus bicipitalis medialis*
Runs along the humerus in the medial bicipital groove

Relationer
Relations

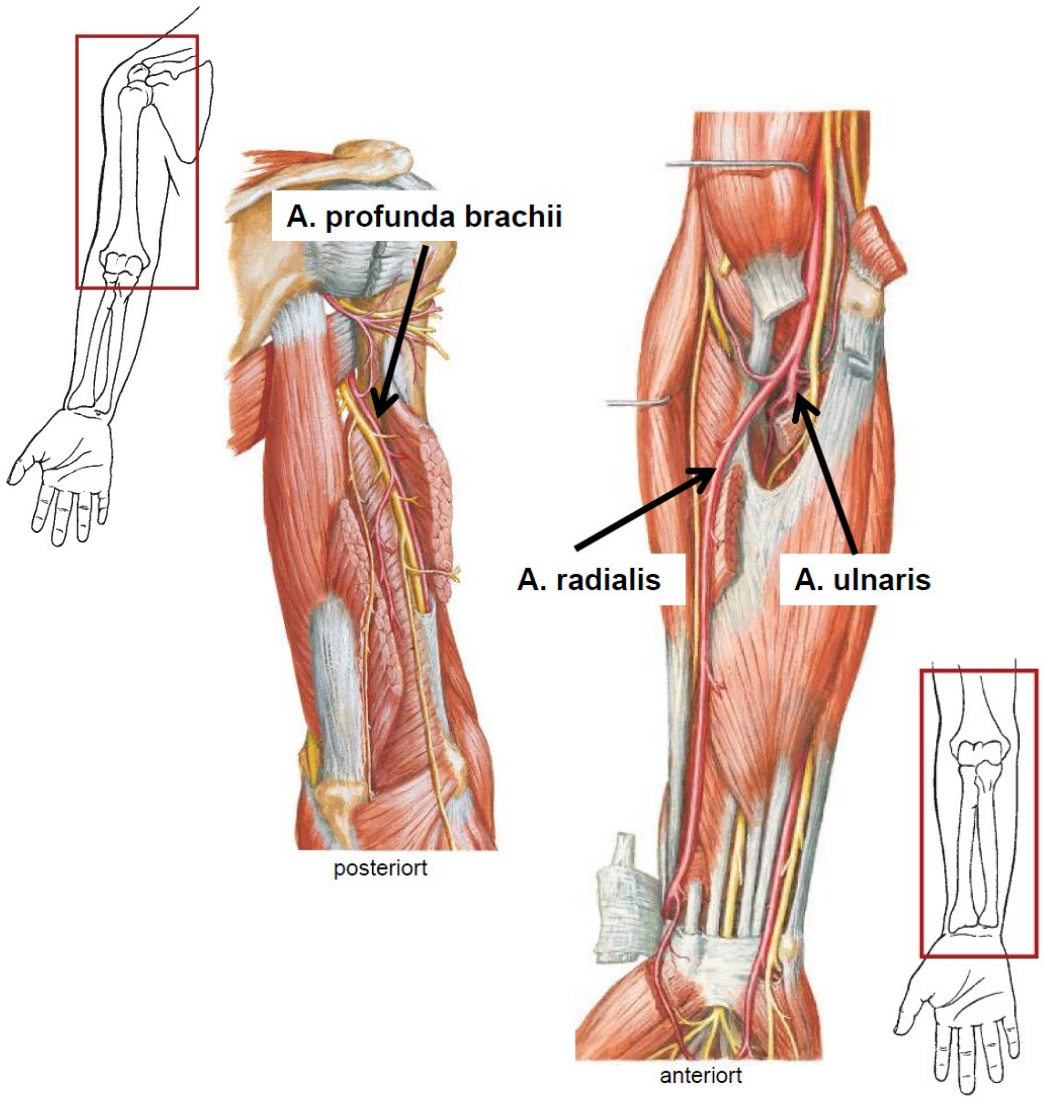
- ***N. ulnaris***
  Ulnar nerve
- ***N. medianus***
  Median nerve
- ***Vv. brachiales***
  Brachial veins
- ***N. cutaneus antebrachii medialis***
  Medial cutaneous nerve of the forearm
- ***N. cutaneus brachii medialis***
  Medial cutaneous nerve of the arm

Brachium (overarmen)

Upper arm

A. brachialis

Brachial artery



Afgiver én vigtig gren: a. profunda brachii
Gives off one important branch: the deep artery of the arm

A. profunda brachii løber til bagsiden sammen med n. radialis
The deep artery of the arm passes posteriorly together with the radial nerve

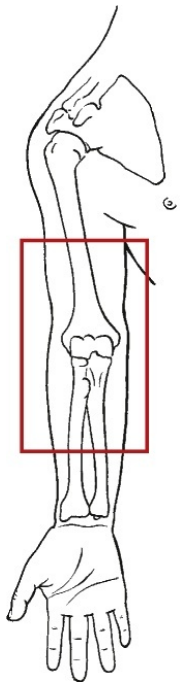
Deling i fossa cubitalis
Division in the cubital fossa

A. brachialis deler sig i a. radialis og a. ulnaris
The brachial artery divides into the radial and ulnar arteries

- A. radialis → over m. pronator teres
  Radial artery → passes over the pronator teres muscle
- A. ulnaris → under m. pronator teres
  Ulnar artery → passes beneath the pronator teres muscle

Fossa cubitalis

Cubital fossa



- **Trekantet rum beliggende foran albueleddet**
- **Afgrænsninger:**

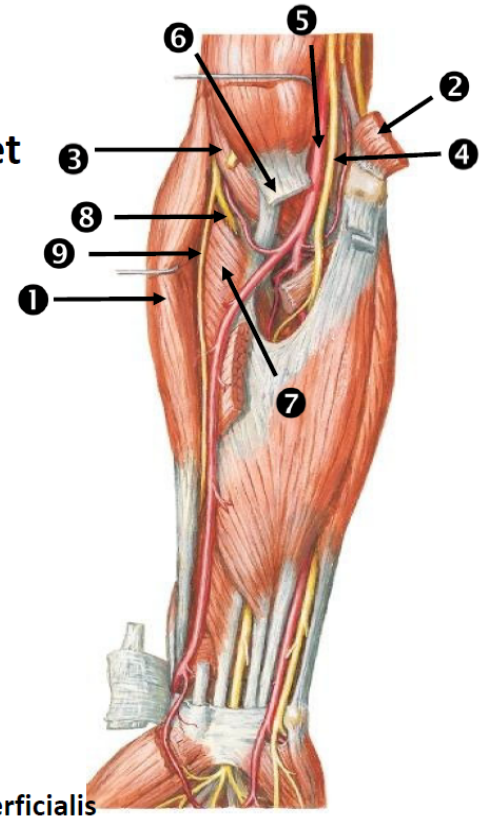
Distalt:
- ❶ M. brachioradialis
- ❷ M. pronator teres

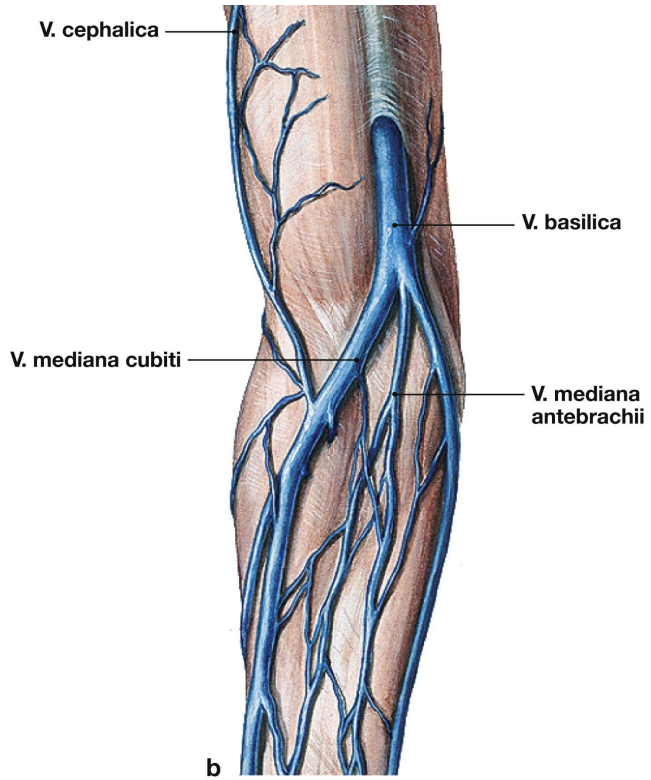
Gulv:
- ❸ M. brachialis
- ❼ M. supinator

Loft
- Fascia antebrachii

Husk at v. mediana cubiti ligger superficielt for fascien

Indhold – nævnt fra medialt til lateralt
- ❹ N. medianus
- ❺ A. brachialis
- ❻ Bicepssenen
- To endegrene fra n. radialis → ❽ r. profundus og ❾ r. superficialis



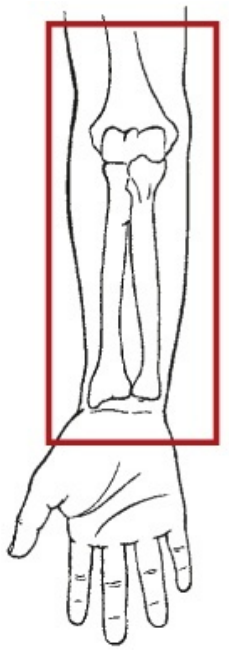


- ***V. mediana cubiti*** **bruges ofte til venepunktur og blodprøvetagning**
  The median cubital vein is commonly used for venipuncture and blood sampling

- ***A. brachialis*** **kan palperes i fossa cubitalis**
  The brachial artery can be palpated in the cubital fossa

Underarm og hånd

Forearm and hand



N. ulnaris

Ulnar nerve

Løber over *retinaculum flexorum*

Passes superficial to the flexor retinaculum

Deler sig i endegrene

Divides into terminal branches

- ***Ramus superficialis***

  Superficial branch

  Kutan

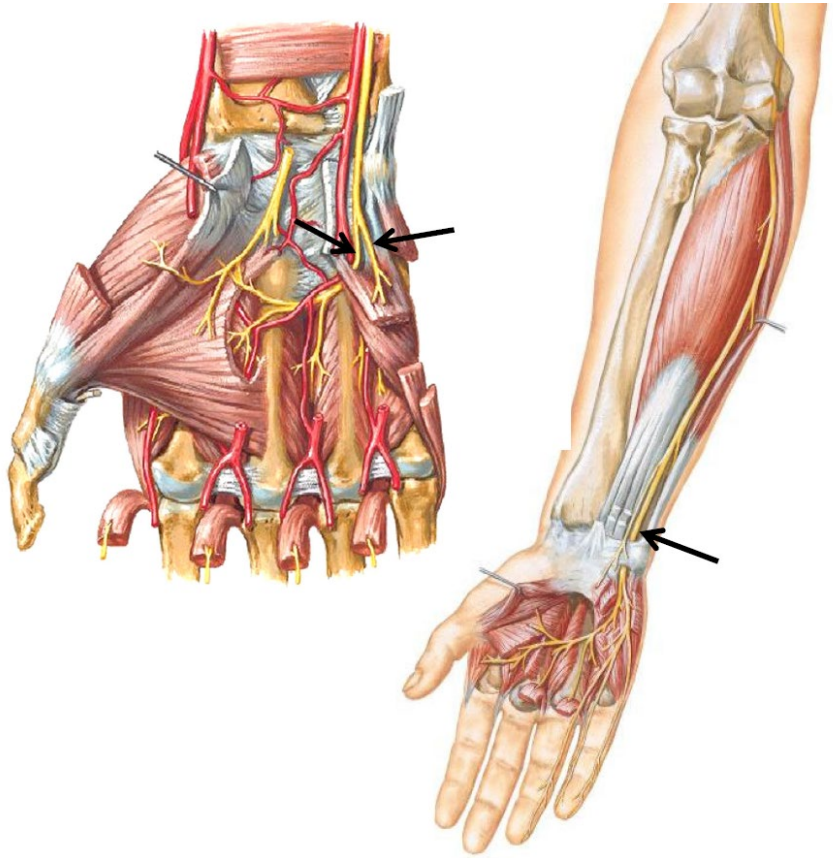
  Sensory / cutaneous

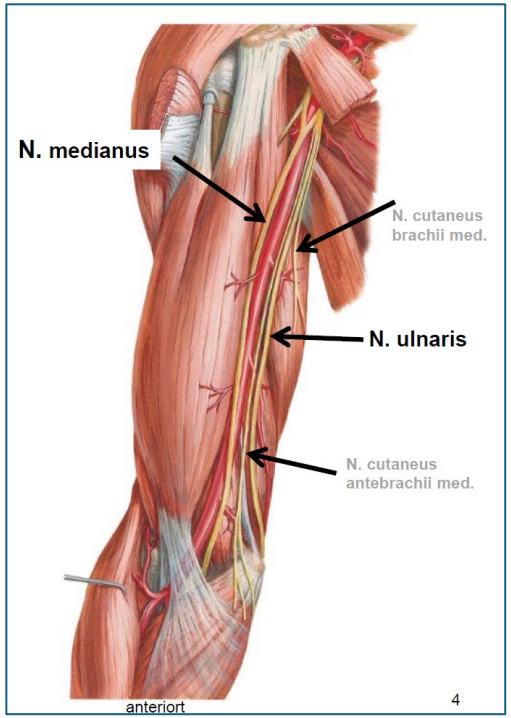
- ***Ramus profundus***

  Deep branch

  Rent muskulær

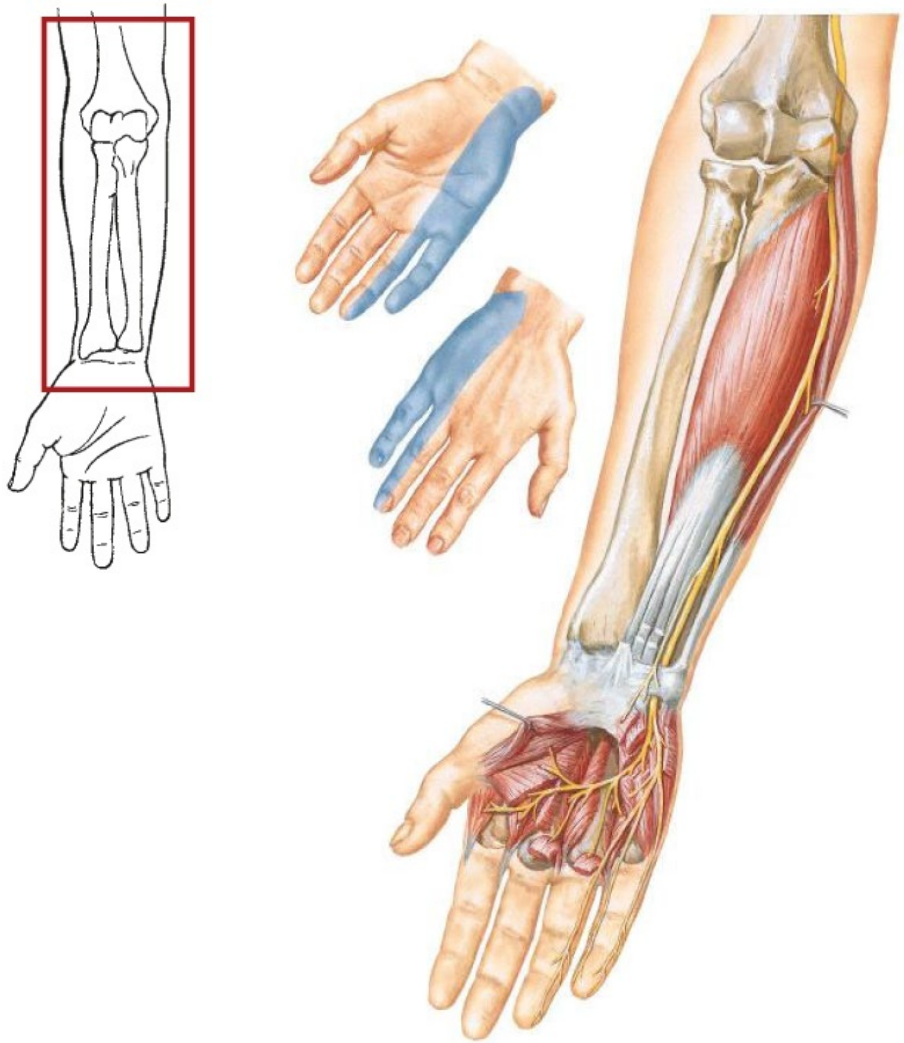
  Purely motor / muscular





Underarm og hånd
Forearm and hand

N. ulnaris
Ulnar nerve



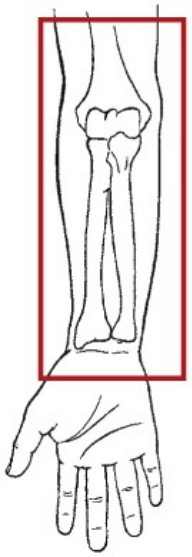
Muskler
Motor innervation
- ***M. flexor carpi ulnaris***
  Flexor carpi ulnaris muscle
- **Ulnare halvdel af *m. flexor digitorum profundus***
  Ulnar half of the flexor digitorum profundus

I hånden innerverer *n. ulnaris* næsten alle intrinsiske håndmuskler
In the hand, the ulnar nerve innervates almost all intrinsic hand muscles
Undtagelser:
Exceptions:
- ***M. abductor pollicis brevis***
- ***M. flexor pollicis brevis***
- ***M. opponens pollicis***
- ***Mm. lumbricales I–II***

Kutant
Cutaneous / sensory innervation
- **Dorsalt: ulnare 1½ fingre**
  Dorsally: the ulnar 1½ fingers
- **Palmart: ulnare 1½ fingre**
  Palmarly: the ulnar 1½ fingers

Underarm og hånd
Forearm and hand

A. ulnaris
Ulnar artery



Udspring og forløb
Origin and course

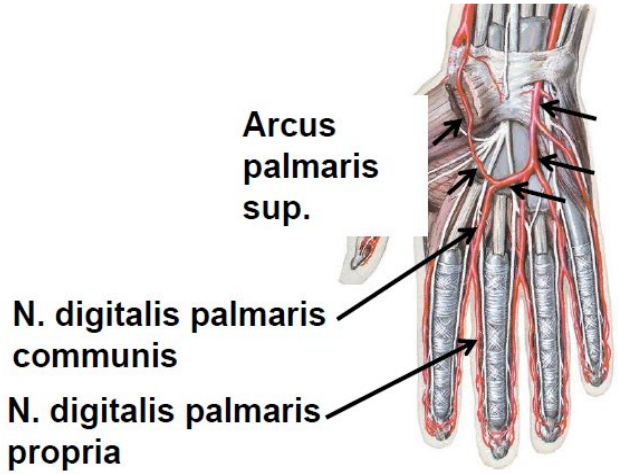
Afgår fra *a. brachialis* i *fossa cubitalis*
Arises from the brachial artery in the cubital fossa

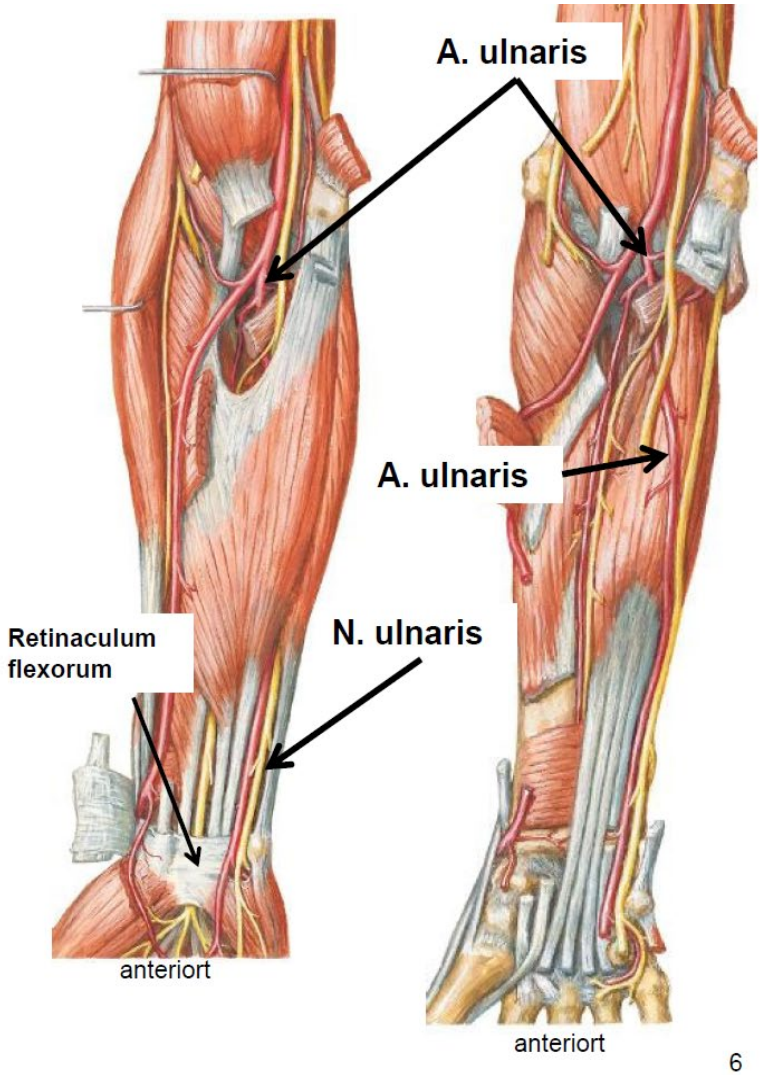
Løber under *m. pronator teres* og senebuen fra *m. flexor digitorum superficialis*
Passes deep to the pronator teres and the tendinous arch of the flexor digitorum superficialis

Løber under *m. flexor carpi ulnaris* og følges af *n. ulnaris*
Runs deep to the flexor carpi ulnaris and is accompanied by the ulnar nerve

Passerer over *retinaculum flexorum*
Passes superficial to the flexor retinaculum

Danner *arcus palmaris superficialis*
Contributes to the superficial palmar arch





Underarm og hånd
Forearm and hand

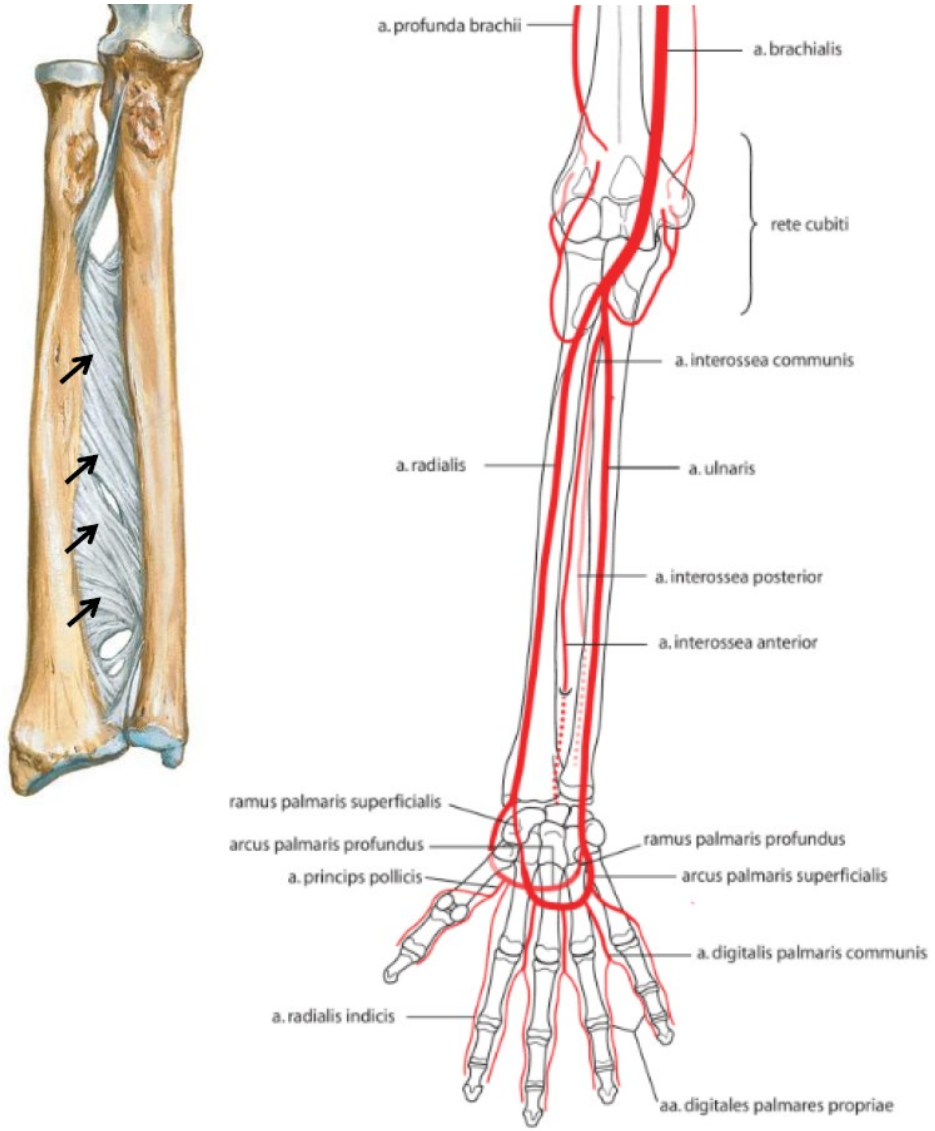
A. ulnaris
Ulnar artery

Afgiver *a. interossea communis*
Gives off the common interosseous artery

- ***A. interossea anterior***
  Anterior interosseous artery
- ***A. interossea posterior***
  Posterior interosseous artery

  ***A. interossea posterior* er den primære blodforsyning til ekstensorlogen**
  The posterior interosseous artery is the main arterial supply to the extensor compartment

- **Afgiver *ramus palmaris profundus* til *arcus palmaris profundus***
  Gives off the deep palmar branch to the deep palmar arch



Underarm og hånd
Forearm and hand

N. radialis
Radial nerve

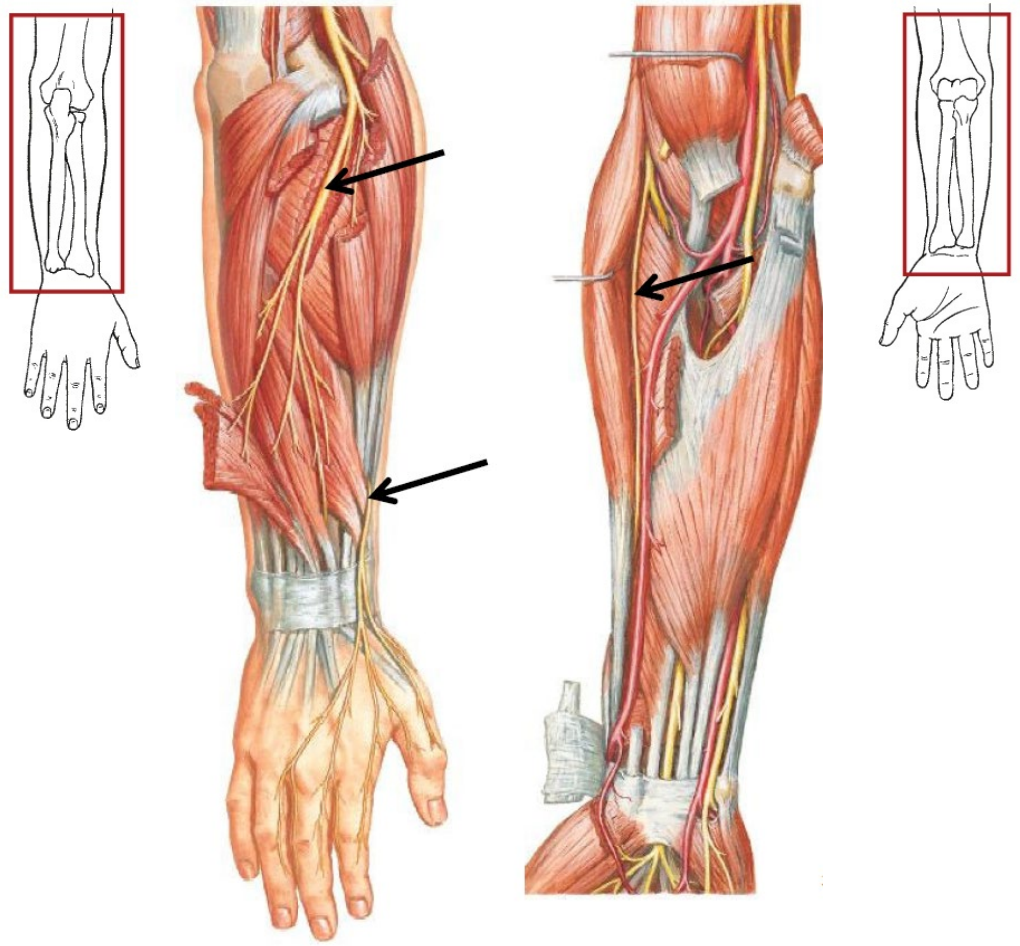
Deler sig i *fossa cubitalis* i to endegrene
Divides in the cubital fossa into two terminal branches

Ramus profundus
Deep branch

- **Går gennem *m. supinator* til ekstensorerne**
  Passes through the supinator muscle to the extensor compartment

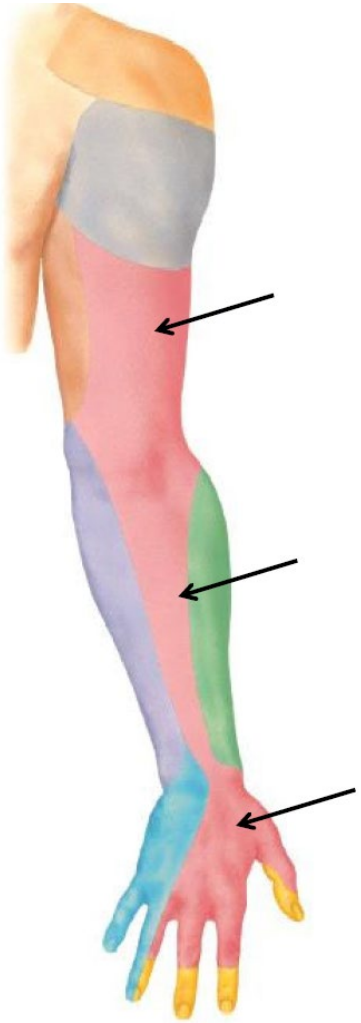
Ramus superficialis
Superficial branch

- **Følger *m. brachioradialis* og *a. radialis***
  Runs with the brachioradialis muscle and radial artery
- **Træder distalt om på *dorsum manus***
  Passes distally onto the dorsum of the hand
- **Kutan innervation: radiale 3½ fingre dorsalt, undtagen neglelejerne**
  Cutaneous innervation: dorsal aspect of the radial 3½ digits, excluding the nail beds



Underarm og hånd
Forearm and hand

N. radialis
Radial nerve



Muskler
Motor innervation

- **Alle ekstensorer i overarmen**
  All extensor muscles of the arm
- **Alle ekstensorer i underarmen**
  All extensor muscles of the forearm

Kutant
Cutaneous / sensory innervation

- **Nederste laterale del af overarmen**
  Lower lateral part of the arm
- **Bagsiden af overarmen**
  Posterior aspect of the arm
- **Bagsiden af underarmen**
  Posterior aspect of the forearm
- **Hånden: proximale dorsale del af de radiale 3½ fingre, undtagen neglelejerne**
  Hand: proximal dorsal aspect of the radial 3½ digits, excluding the nail beds

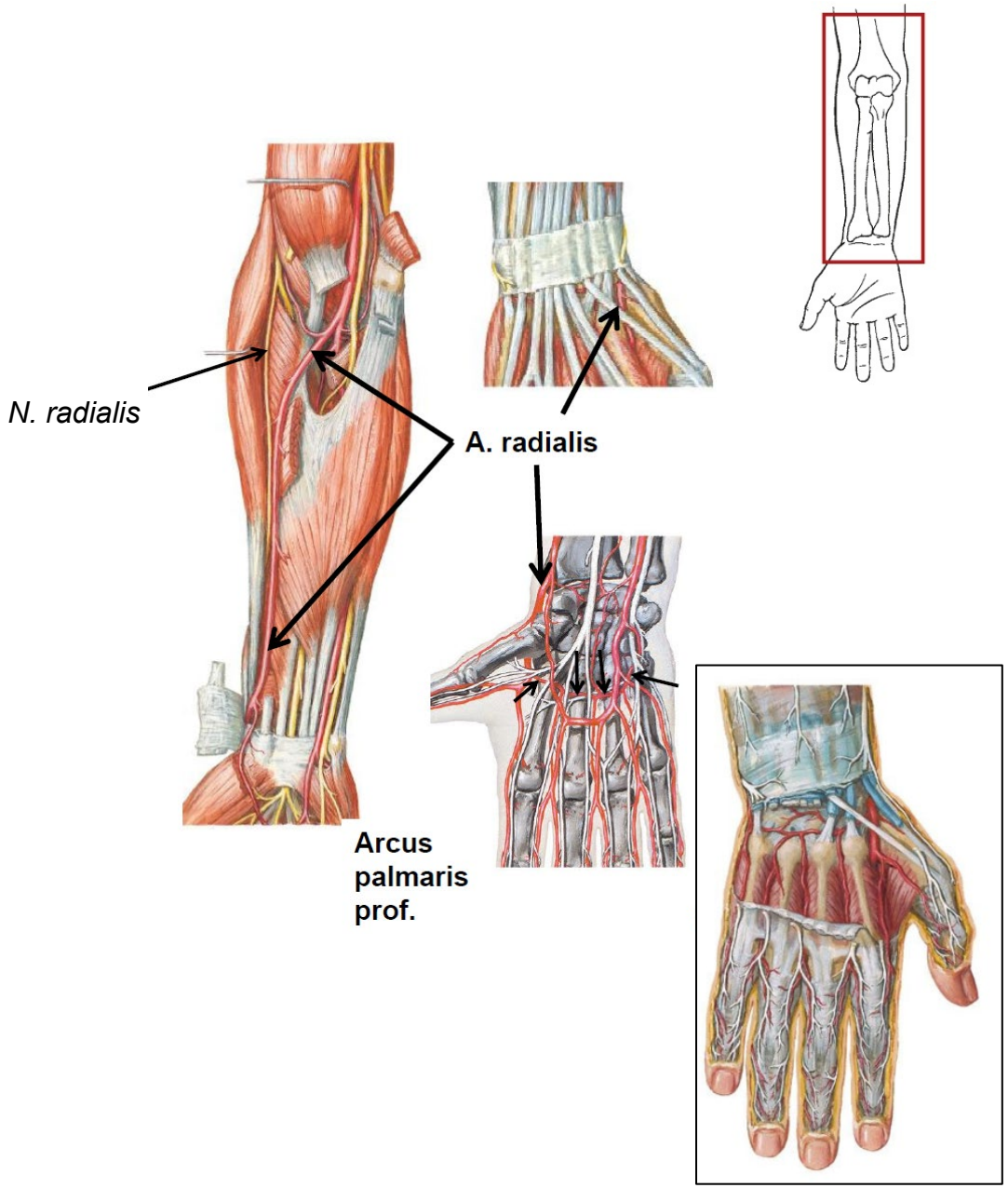
Underarm og hånd
Forearm and hand

A. radialis
Radial artery

Udspring og forløb
Origin and course

Fra *a. brachialis* i *fossa cubitalis*
Arises from the brachial artery in the cubital fossa

- Løber over *m. pronator teres*
  Runs superficial to the pronator teres muscle
- Løber langs *m. brachioradialis* sammen med *r. superficialis n. radialis*
  Runs along the brachioradialis muscle together with the superficial branch of the radial nerve
- Ligger superficielt distalt på radius → her kan pulsen palperes
  Lies superficially over the distal radius → the pulse can be palpated here
- Løber dorsalt til den anatomiske tabatière
  Passes dorsally through the anatomical snuffbox
- Håndens arterielle forbindelse
  Arterial connection in the hand
  - *Går ind i palma gennem m. interosseus dorsalis I*
    Enters the palm by passing through the first dorsal interosseous muscle
  - *Danner arcus palmaris profundus*
    Forms the deep palmar arch



Underarm og hånd
Forearm and hand

A. radialis
Radial artery

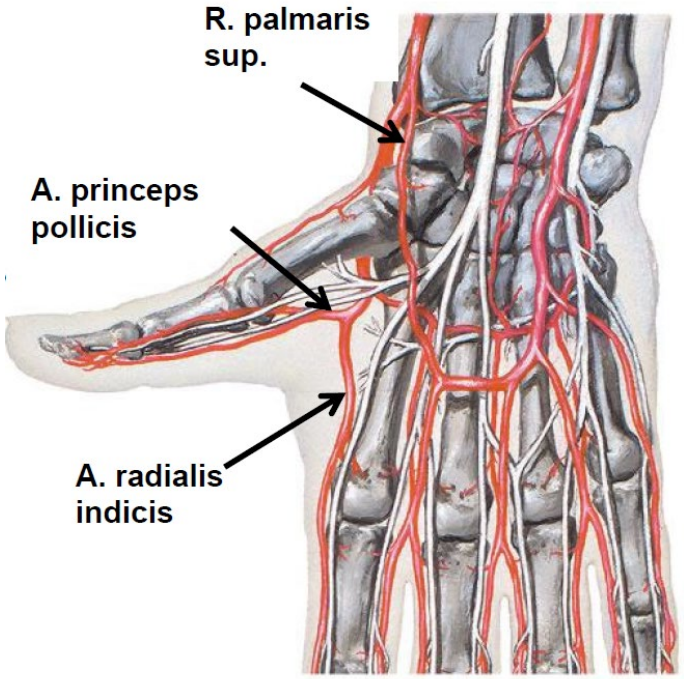
Afgiver grene
Gives off branches

- *Ramus palmaris superficialis*
  Superficial palmar branch
- *A. princeps pollicis*
  Princeps pollicis artery
- *A. radialis indicis*
  Radial artery of the index finger

Arcus palmaris superficialis
Superficial palmar arch

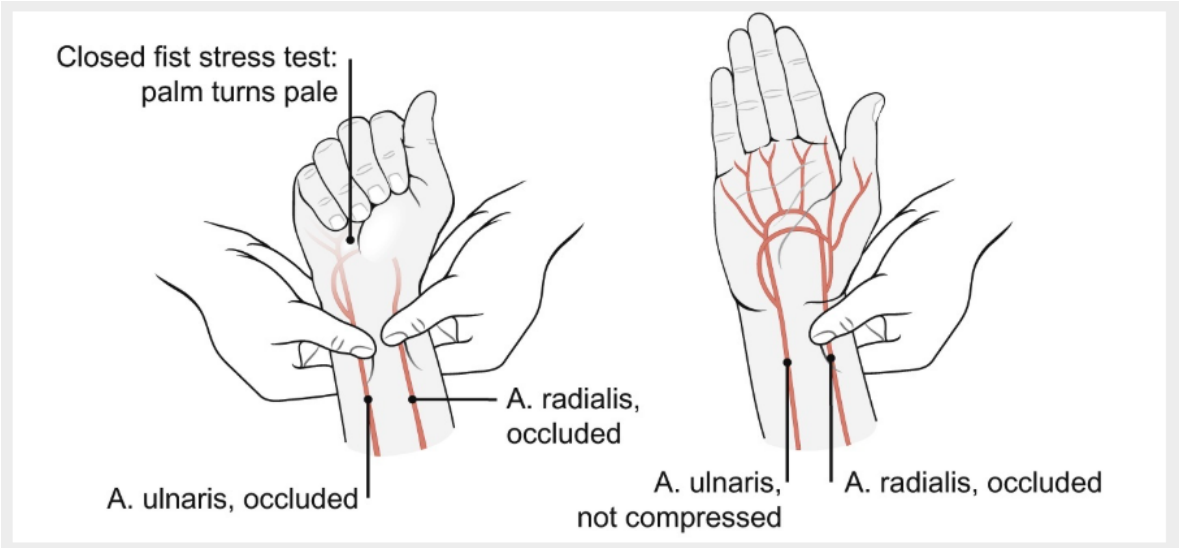
- *ulnaris* + *r. palmaris superficialis* fra *a. radialis*
  Formed mainly by the ulnar artery, with contribution from the superficial palmar branch of the radial artery

Arcus palmaris profundus
Deep palmar arch
A*. radialis* + *r. palmaris profundus* fra *a. ulnaris*
Formed mainly by the radial artery, with contribution from the deep palmar branch of the ulnar artery



Underarm og hånd
Forearm and hand

Allen-test



Kompression af begge arterier → slip a. ulnaris → vurder reperfusion

Compress both arteries → release the ulnar artery → assess reperfusion

Normal test: Hurtig farvetilbagekomst → tilstrækkelig kollateral cirkulation.

Rapid return of colour → adequate collateral circulation.

Hånd
Hand

Canalis carpi
Carpal tunnel

Indgangen til håndens centrale loge
Entrance to the central compartment of the hand

Afgrænsninger
Boundaries

- Karpalknoglerne + ligamenter
  Carpal bones and ligaments

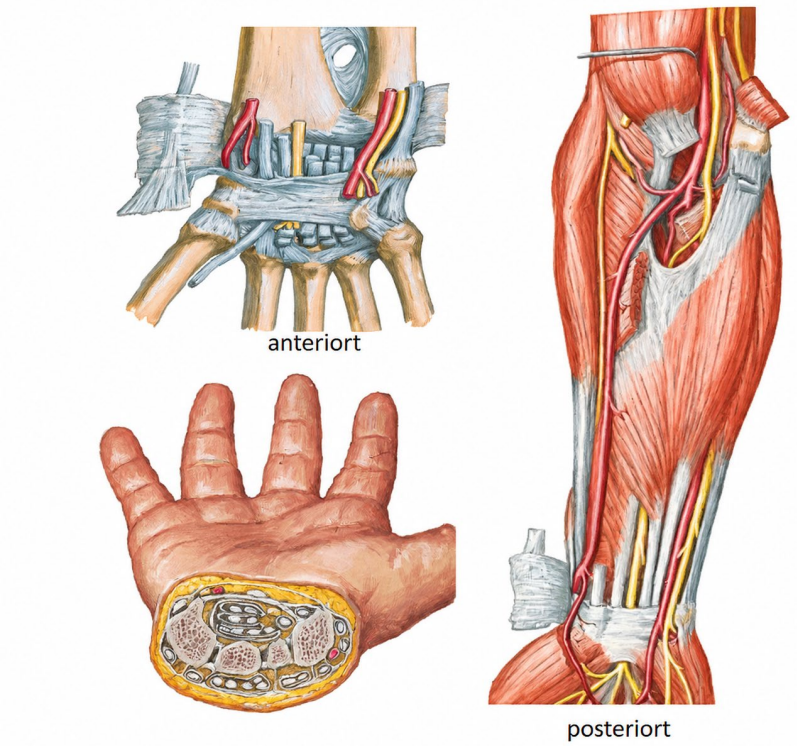
- *Retinaculum flexorum*
  Flexor retinaculum

Indhold

9 fleksorsener
9 flexor tendons

- *M. flexor pollicis longus*
  Flexor pollicis longus tendon
- *M. flexor digitorum superficialis*
  Flexor digitorum superficialis tendons
- *M. flexor digitorum profundus*
  Flexor digitorum profundus tendons

N. medianus
Median nerve



Regional anatomy becomes clinically useful when you understand the relationships between structures.

1
What forms the space?
Boundaries and landmarks

2
What passes through it?
Nerves, vessels and tendons

3
What lies next to what?
Anatomical relationships